Ihara et al., 2021

Handwriting vs typing: learning new words

Writing new words **by hand** (ink pen or digital pen) left a **bigger brain trace of learning** than typing, even though typing let people write **23% more repetitions**. Test scores right after did not differ.

The study

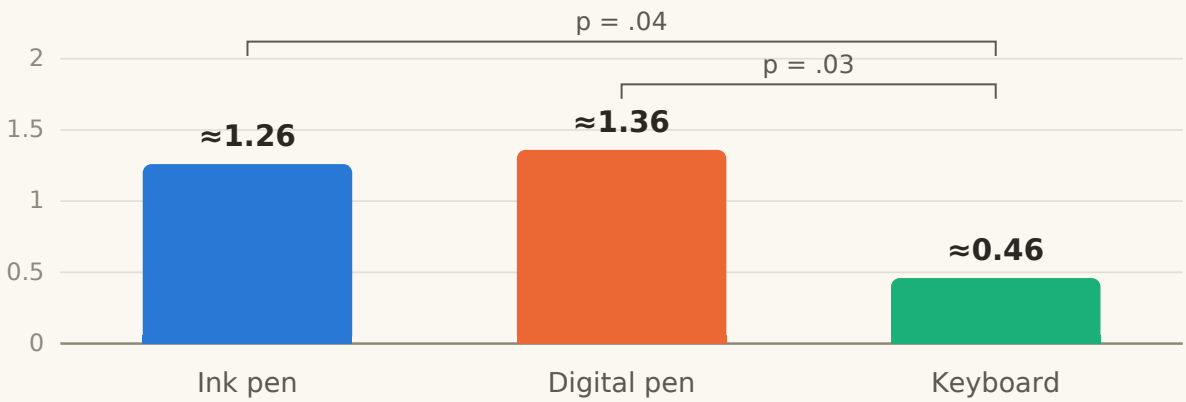
33	60	10 min
adults (Japanese)	Indonesian words	per tool, 20 words

Learn → **Mood check** → **Word test** → **EEG (N400)**

Ink pen · Digital pen · Keyboard

Brain learning signal

N400 priming effect, μV ≈ · bigger = more learning



...but typing gave more practice

writing repetitions per word



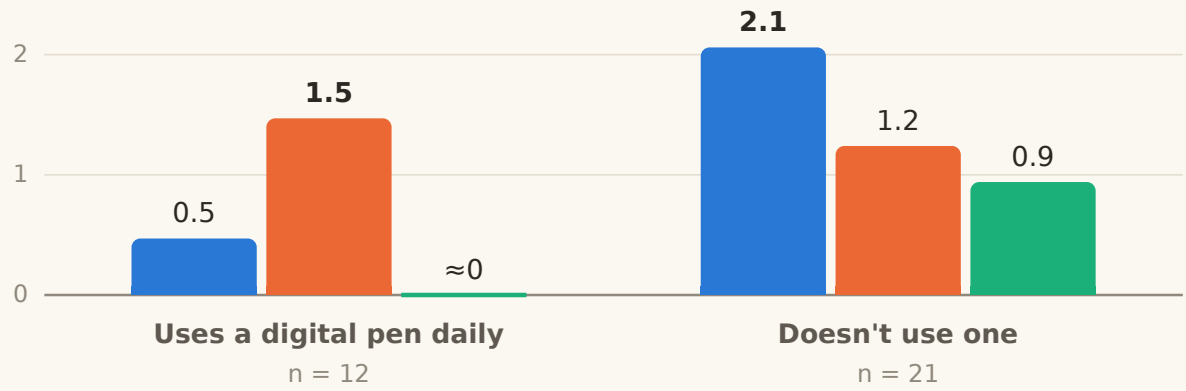
More repetitions ≠ more learning.

keyboard > both pens, p < .05

Ink pen Digital pen Keyboard

Familiarity with the digital pen matters

N400 size, μV ≈



Used to a digital pen: the digital pen was highest (vs keyboard p = .02). Not used to it: the ink pen was highest (vs keyboard p = .057, only a trend).

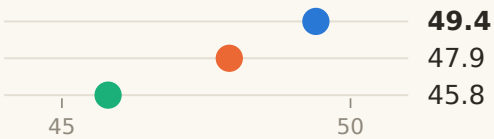
Words remembered

9–12 of 20

for every tool (8.8–11.8)

no difference (p = .19)

Felt energetic (Vigor) ≈



T-score; ink > keyboard p = .036

Read with care

- Small groups (12 and 21 people).
- Measured right after learning only; no long-term test.
- The brain signal differed; test scores did not.
- The digital pen was a Wacom pen display, not an iPad.
- Co-funded by a pen-tablet maker (Wacom); two authors work there.

Миний хувьд

- Шинэ үг цээжлэхдээ гараар бич, бүү шив.
- GoodNotes + Apple Pencil: дассан бол цаас шиг сайн; дасаагүй бол цаасаар бич.
- Олон давтах нь илүү сурсан гэсэн үг биш.

Ihara AS, Nakajima K, Kake A, Ishimaru K, Osugi K, Naruse Y (2021). Advantage of handwriting over typing on learning words: evidence from an N400 event-related potential index. Front. Hum. Neurosci. 15:679191. doi:10.3389/fnhum.2021.679191 · CC BY 4.0. ≈ = measured from the paper's figures.